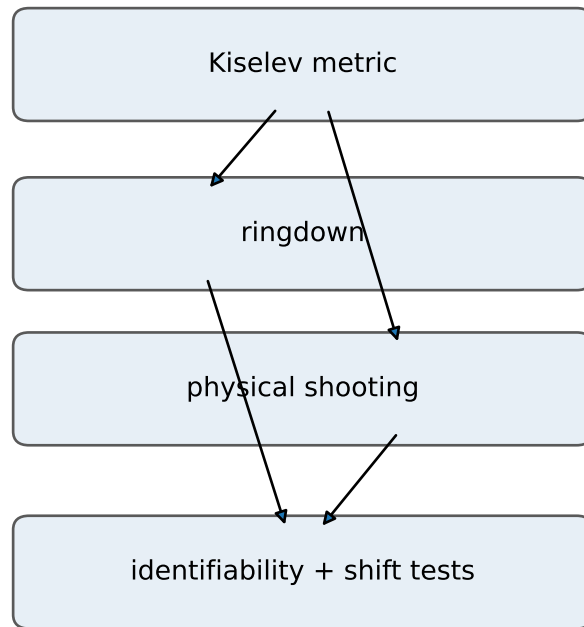
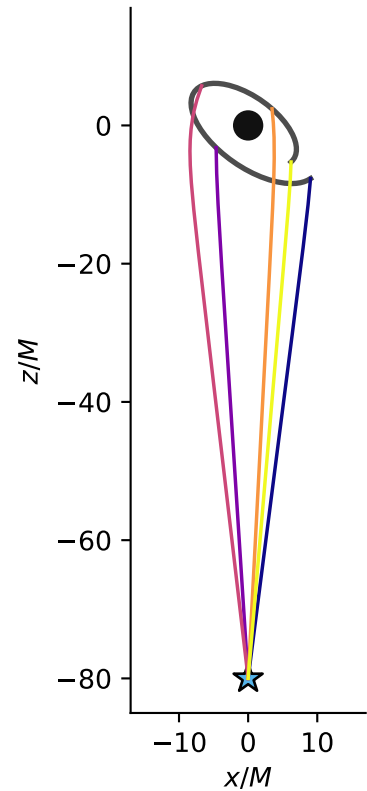


Physical observables reveal when black-hole hair is identifiable

validated direct rays



complementary sky response

